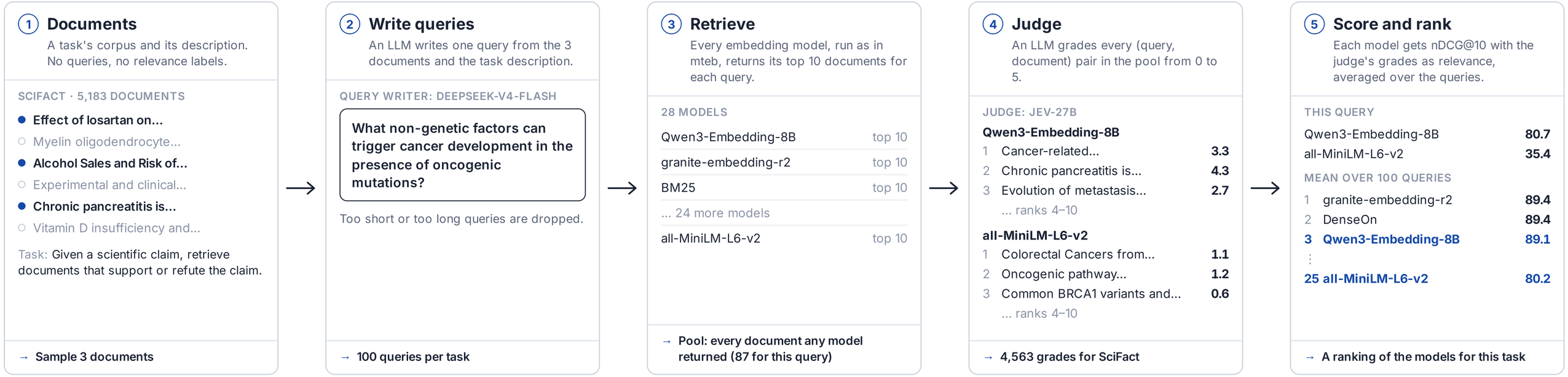

1 Documents
A task's corpus and its description. No queries, no relevance labels.
SCIFACT · 5,183 DOCUMENTS
Effect of losartan on...
Myelin oligodendrocyte...
Alcohol Sales and Risk of...
Experimental and clinical...
Chronic pancreatitis is...
Vitamin D insufficiency and...
Task: Given a scientific claim, retrieve documents that support or refute the claim.
→ Sample 3 documents
2 Write queries
An LLM writes one query from the 3 documents and the task description.
QUERY WRITER: DEEPSEEK-V4-FLASH
What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?
Too short or too long queries are dropped.
→ 100 queries per task
3 Retrieve
Every embedding model, run as in mteb, returns its top 10 documents for each query.
28 MODELS
Qwen3-Embedding-8B top 10
granite-embedding-r2 top 10
BM25 top 10
... 24 more models
all-MiniLM-L6-v2 top 10
→ Pool: every document any model returned (87 for this query)
4 Judge
An LLM grades every (query, document) pair in the pool from 0 to 5.
JUDGE: JEV-27B
Qwen3-Embedding-8B
1 Cancer-related... 3.3
2 Chronic pancreatitis is... 4.3
3 Evolution of metastasis... 2.7
... ranks 4–10
all-MiniLM-L6-v2
1 Colorectal Cancers from... 1.1
2 Oncogenic pathway... 1.2
3 Common BRCA1 variants and... 0.6
... ranks 4–10
→ 4,563 grades for SciFact
5 Score and rank
Each model gets nDCG@10 with the judge's grades as relevance, averaged over the queries.
THIS QUERY
Qwen3-Embedding-8B 80.7
all-MiniLM-L6-v2 35.4
MEAN OVER 100 QUERIES
1 granite-embedding-r2 89.4
2 DenseOn 89.4
3 Qwen3-Embedding-8B 89.1
25 all-MiniLM-L6-v2 80.2
→ A ranking of the models for this task